Theorem. (Erdős #152) For a Sidon set $A \subset \mathbb{N}$ of size n, let $I(A+A)$ count the isolated elements of the sumset — those $s \in A+A$ with $s \pm 1 \notin A+A$. Define $f(n)$ as the minimum of $I(A+A)$ over all Sidon sets of size n. Then $f(n) \geq (n^2 - 100n - 16)/16$, so in particular $f(n) \to \infty$.
Proof. The proof establishes, for every Sidon set A of size n:

$$16 \cdot I(A+A) + 100n + 16 \geq n^2. \qquad (\star)$$

Let $D = \{a - b : a, b \in A\}$ denote the difference set and $S = A + A$ the sumset. For any set $X \subseteq \mathbb{Z}$, let $N_k(X) := |x \in X : x + k \in X|$ be the number of pairs at distance k, $V_2(X) := |x \in X : x \pm 1 \in X|$ be the number of interior points, and $I(X)$ be the isolated points .

Let $X \subset \mathbb{Z}$. We will employ the following three facts:

1. $I(X) + 2N_1(X) = |X| + V_2(X)$, by partitioning elements by neighbor count.

2. $4N_1(X) + N_3(X) \leq 3|X| + 2N_2(X)$, which follows from writing $N_k(x) = \sum_{x \in \mathbb{Z}} 1_X(x) 1_X(x+k)$ and a pointwise check of the indicator function 1_X across all 4-point windows $(x, x+1, x+2, x+3)$; that is, summing the local inequality $1_X(x) + 1_X(x+1) + 1_X(x+2) + 1_X(x)1_X(x+2) + 1_X(x+1)1_X(x+3) \geq 1_X(x)1_X(x+1) + 21_X(x+1)1_X(x+2) + 1_X(x+2)1_X(x+3) + 1_X(x)1_X(x+3)$ and then summing over all $x \in \mathbb{Z}$.

3. $2N_2(X) \leq N_3(X) + 2V_2(X) + 2I(X)$. Consider each pair $(x, x+2) \in X^2$ and we proceed by cases on $x+1$. Let $G = \{x \in X \mid x+1 \notin X \wedge x+2 \in X\}$. If $x+1 \in X$, then the pair is counted exactly by V_2, so $N_2(X) = V_2(X) + |G|$. If $x+1 \notin X$, then

   - If $x - 1 \in X$, then $(x-1, x+2)$ is a pair counted in $N_3(X)$.
   - If $x + 3 \in X$, then $(x, x+3)$ is a pair counted in $N_3(X)$.
   - If $x - 1 \notin X$, then x is an isolated point in $I(X)$, and if $x+3 \notin X$, then $x+2$ is an isolated point in $I(X)$.

   Since each N_3 pair can be counted by two different gaps, while isolated points are distinct in this construction, we have $2|G| \leq N_3(X) + 2I(X)$.

Next, for each $k \geq 1$, count quadruples $(a, b, c, d) \in A^4$ with $a + b + k = c + d$. Rewriting as $a - c = d - b - k$ and partitioning by $\delta = a - c$ yields the following two quadruple transfer bounds:

4. The cases $\delta = 0$ and $\delta = -k$ each contribute $\leq |A|$ (one coordinate determines the rest). For any $\delta \notin \{0, -k\}$, the Sidon property ensures that there is at most one pair (a, c) with $a - c = \delta$ and at most one pair (d, b) with $d - b = \delta + k$. Consequently, each gap of size k in D identifies exactly one quadruple. Thus $|\text{quad}_k| \leq N_k(D) + 2n$.

5. Each "good" element $s \in S$ with $s + k \in S$ (excluding $\leq 2n$ doubles of the form $2a$) produces ≥ 4 quadruples, giving $4N_k(S) \leq |\text{quad}_k| + 8n$.

Combining these bounds shows that $N_k(D)$ and $N_k(S)$ are related up to $O(n)$ error. Applying (2) to $D = A - A$ gives

$$4N_1(D) + N_3(D) \leq 3|D| + 2N_2(D)$$

and then substitute the quadruple transfer bounds (4, 5) for $k = 1, 2, 3$ and collecting all $O(n)$ terms gives

$$16N_1(S) + 4N_3(S) \leq 3|D| + 8N_2(S) + 100n$$

Then, using fact (3) on S we obtain

$$16N_1(S) + 4N_3(S) \leq 3|D| + (4N_3(S) + 8V_2(S) + 8I(S)) + 100n$$

and notice that the $4N_3(S)$ terms cancel. Then applying fact (1) we obtain

$$(8|S| + 8V_2(S) - 8I(S)) \leq 3|D| + 8V_2(S) + 8I(S) + 100n$$

and see that the $8V_2(S)$ terms now cancel, leaving

$$16I(S) + 100n \geq 8|S| - 3|D|.$$

The standard Sidon estimates $|S| \geq n^2/2$ and $|D| \leq n^2$ then give $16I(S) + 100n \geq 4n^2 - 3n^2 = n^2$, accounting for a boundary correction of $+16$ at zero yields $(\star)$. $\square$